



PAPER ID-310607

Printed Page: 1 of 3
Subject Code: BMBAI304

Roll No:

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MBA-INT
(SEM III) THEORY EXAMINATION 2025-26
QUANTITATIVE TECHNIQUES

TIME: 3 HRS**M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

02 x 7 = 14

Q no.	Question	CO	Level
a.	List the application of operation research model.	1	K2
b.	When do players apply mixed strategies?		
c.	What is total float?		
d.	What is the standard form of LP Problem?		
e.	List out the methods used to obtain initial basic feasible solution of a transportation problem.		
f.	Define the term Reneging in queuing theory.		
g.	Why does dummy activity are used in CPM?		

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

a.	State any four areas for the application of OR techniques in financial management. How do they improve the performance of an organization?	1	K2																			
b.	Use the graphical method to solve the following LP problem. Maximize $Z = 4x_1 + 4x_2$ Subject to $x_1 + 2x_2 \leq 10$ $6x_1 + 6x_2 \leq 36$ $x_1 \leq 6$ And $x_1, x_2 \geq 0$																					
c.	Discuss Hungarian method to find the solution of assignment problem..																					
d.	A company has three options to choose against the three events. The payoff matrix of the problem is given below : <table border="1" style="margin: 10px auto;"> <thead> <tr> <th rowspan="2">State of Nature</th><th colspan="3">Options</th></tr> <tr> <th>O₁</th><th>O₂</th><th>O₃</th></tr> </thead> <tbody> <tr> <td>E₁</td><td>25</td><td>-10</td><td>125</td></tr> <tr> <td>E₂</td><td>400</td><td>440</td><td>4000</td></tr> <tr> <td>E₃</td><td>650</td><td>740</td><td>750</td></tr> </tbody> </table> The probabilities of the states of nature are 0.1, 0.7 and 0.2 respectively. Calculate and tabulate the EMV and conclude which would prove to be the best course of action.	State of Nature	Options			O ₁	O ₂	O ₃	E ₁	25	-10	125	E ₂	400	440	4000	E ₃	650	740	750		
State of Nature	Options																					
	O ₁	O ₂	O ₃																			
E ₁	25	-10	125																			
E ₂	400	440	4000																			
E ₃	650	740	750																			
e.	Explain the graphical method to solve two jobs N machines sequencing problem with given technological ordering of each jobs.																					

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

3. Attempt any one part of the following.			
a.	Discuss the concept of decision tree analysis and its components with suitable examples.	1	K2
b.	A farmer wants to decide which of the three crops he should plant on his 100-acre form. The profit from each is dependent on the rainfall during the growing season. The farmer has to categorize the amount of rainfall as high, medium and low. His estimated profit for each is shown in the table below : Estimated conditional profit (Rs.) Crop A Crop B Crop C		

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High	8000	3500	5000
Medium	4500	4500	5000
Low	2000	5000	4000

If the farmer wishes to plant only one crop, decide which should be his "best crop" using:

- Maximax criterion.
- Maximin criterion.
- Hurwicz criterion.
- Laplace criterion.
- Minimax regret criterion.

4. Attempt any *one* part of the following:

07 x 1 = 07

- | | | | |
|----|---|---|----|
| a. | Describe the Maximin and minimax principal in game theory. What do you understand by two-person zero-sum game, pure strategies, mixed strategies and saddle point in game theory? | 1 | K2 |
| b. | Find the optimal assignment for the following profit matrix. | | |

5. Attempt any *one* part of the following.

07 11 = 07

- | a. | <p>Determine an optimal feasible solution to the following transportation problem using MODI method.</p> <table border="1" data-bbox="416 1232 1216 1281"> <tr> <th rowspan="2">Source</th><th colspan="5">Destination</th></tr> <tr> <th>D1</th><th>D2</th><th>D3</th><th>D4</th><th>Supply</th></tr> <tr> <td>S1</td><td>11</td><td>13</td><td>17</td><td>14</td><td>250</td></tr> <tr> <td>S2</td><td>16</td><td>18</td><td>14</td><td>10</td><td>300</td></tr> <tr> <td>S3</td><td>21</td><td>24</td><td>13</td><td>10</td><td>300</td></tr> <tr> <td>Demand</td><td>200</td><td>225</td><td>275</td><td>200</td><td></td></tr> </table> | | | | | | Source | Destination | | | | | D1 | D2 | D3 | D4 | Supply | S1 | 11 | 13 | 17 | 14 | 250 | S2 | 16 | 18 | 14 | 10 | 300 | S3 | 21 | 24 | 13 | 10 | 300 | Demand | 200 | 225 | 275 | 200 | |
|--------|--|-----|-----|-----|--------|--|--------|-------------|--|--|--|--|----|----|----|----|--------|----|----|----|----|----|-----|----|----|----|----|----|-----|----|----|----|----|----|-----|--------|-----|-----|-----|-----|--|
| Source | Destination | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | D1 | D2 | D3 | D4 | Supply | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S1 | 11 | 13 | 17 | 14 | 250 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S2 | 16 | 18 | 14 | 10 | 300 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| S3 | 21 | 24 | 13 | 10 | 300 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Demand | 200 | 225 | 275 | 200 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b. | <p>What does you mean by degenerated transportation problem? Write the procedure to solve the problem of degeneracy.</p> | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

6. Attempt any *one* part of the following:

07 x 1 = 07

- a. Use the graphical method for the following game to find the best strategies of both players and the value of the game.

Player A		Player B			
		B ₁	B ₂	B ₃	B ₄
A ₁		1	3	-1	4
A ₂		-3	5	6	1

b. We have five jobs, each of which must go through the machines A, B and C in the order ABC. Processing time in hours) is as follows.

Jobs	1	2	3	4	5
Machine A	5	7	6	9	5
Machine B	2	1	4	5	3
Machine C	3	7	5	6	7

Determine the optimal sequence, total elapsed time and idle time of each machine.

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7. Attempt any *one* part of the following:

07 x 1 = 07

a. What do you understand by (i) queue discipline, (ii) arrival process and (iii) service process?

1K2

b. The following table gives a list of various activities and their immediate predecessors involved in installation of CAT scanner in a hospital.

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Immediate predecessors	-	A	B	B	C	C	C	E,F,G	H	D,I	J	K	L
Expected duration (days)	2	6	12	8	6	4	3	10	9	6	3	2	1

(a) Prepare a network diagram for the project.

(b) Find the total time required to install CAT scanner.

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